

An Interactive Visualization System to Analyze and Predict Urban Construction Dynamics

SENSORO
Connect the Physical and Digital Worlds



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Why Studying Urban Construction?

- Increased records of **interaction** of city and human activity
 - eg: the 311 call system in the New York City
- Benefit
 - For Government
 - Budget control
 - Resource Allocation
 - For Citizens
 - Better knowing of road construction in neighborhoods
 - Better supervising the government

What is Urban Construction?



Urgent Repair



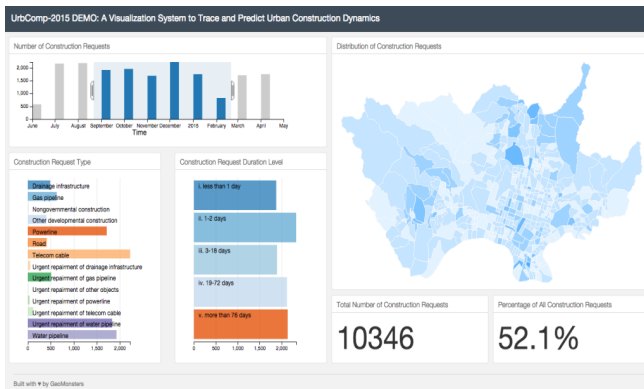
Scheduled Construction

```
1 var dataSchema = {  
2   location: {  
3     latitude: Number,  
4     longitude: Number  
5   },  
6   time: new Date(),  
7   type: String  
8   ...  
9 };
```

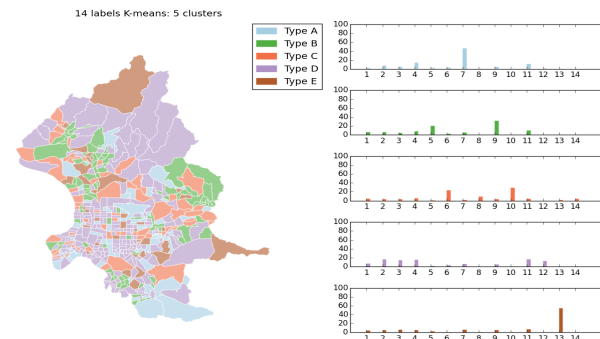
Our Goals

- Develop a system to present, analyze and predict the urban constructions, featuring:

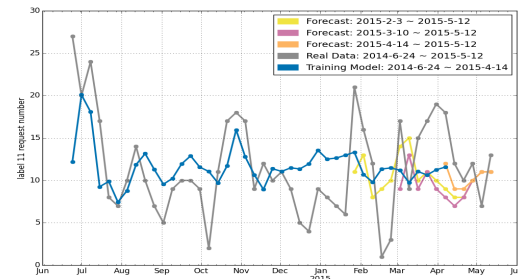
1. Interactive Data Processing and Visualisation



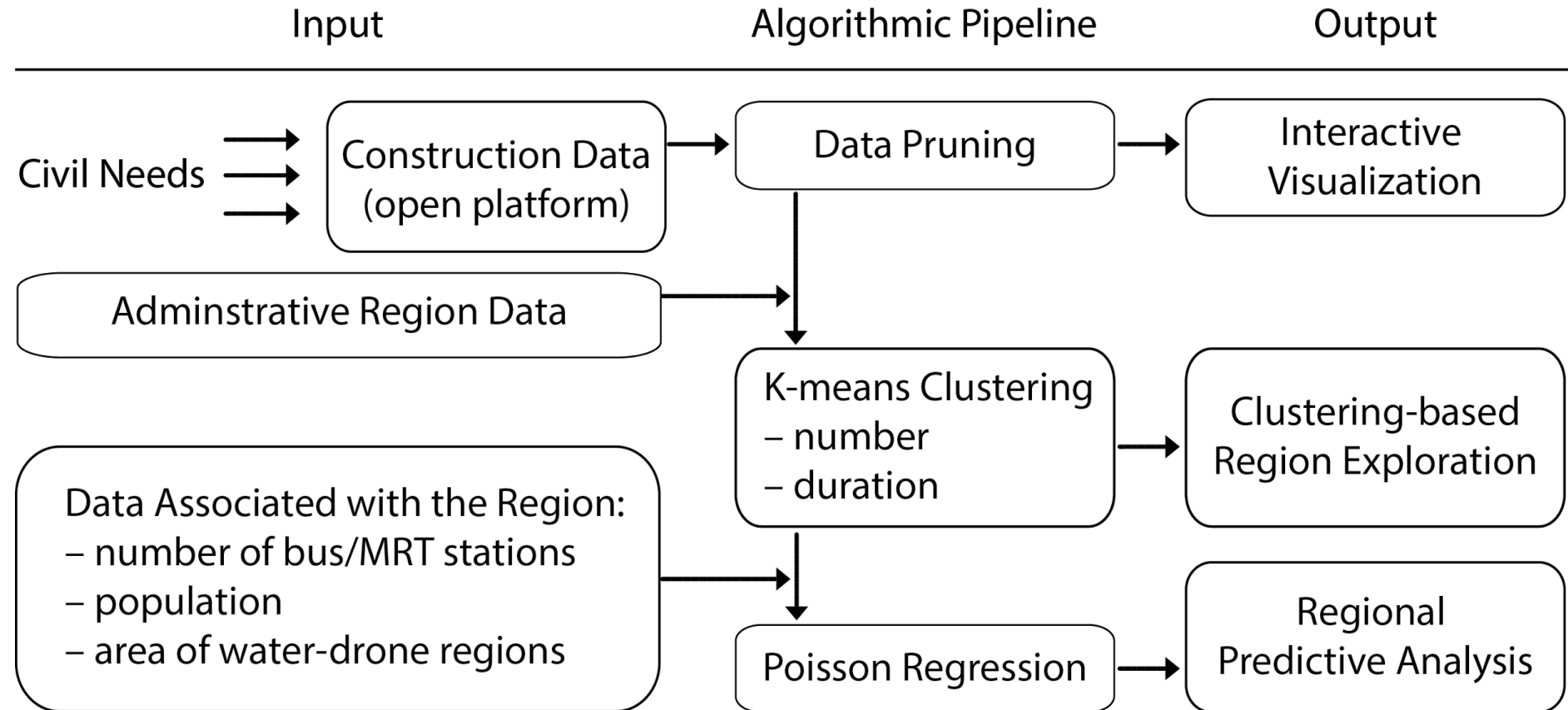
2. Clustering-based Region Exploration



3. Construction Request Forecasting



System Framework



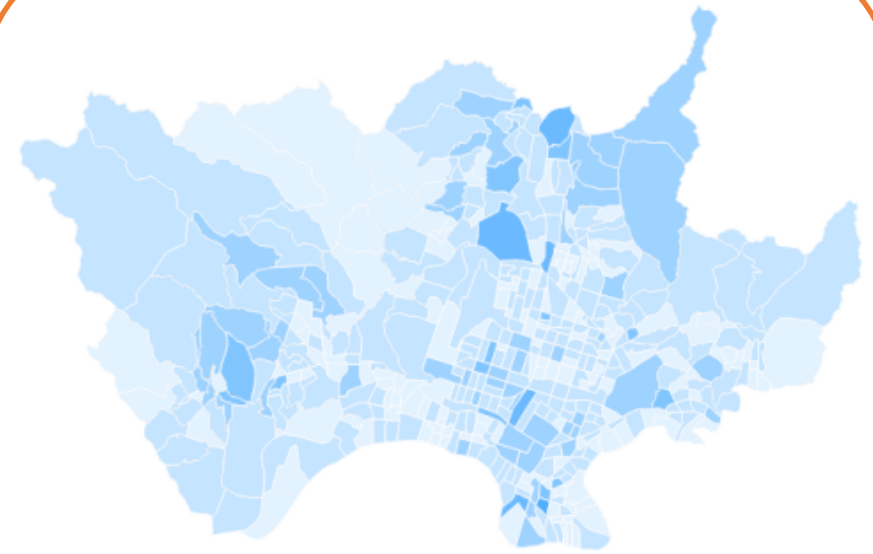
System Snapshot

UrbComp-2015 DEMO: A Visualization System to Trace and Predict Urban Construction Dynamics

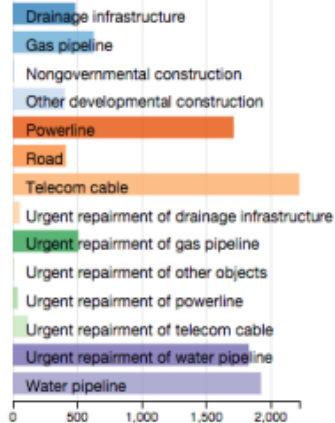
Number of Construction Requests



Distribution of Construction Requests



Construction Request Type



Construction Request Duration Level



Total Number of Construction Requests

10346

Percentage of All Construction Requests

52.1%

Data Pruning

1. Cleaning the outliers from raw data

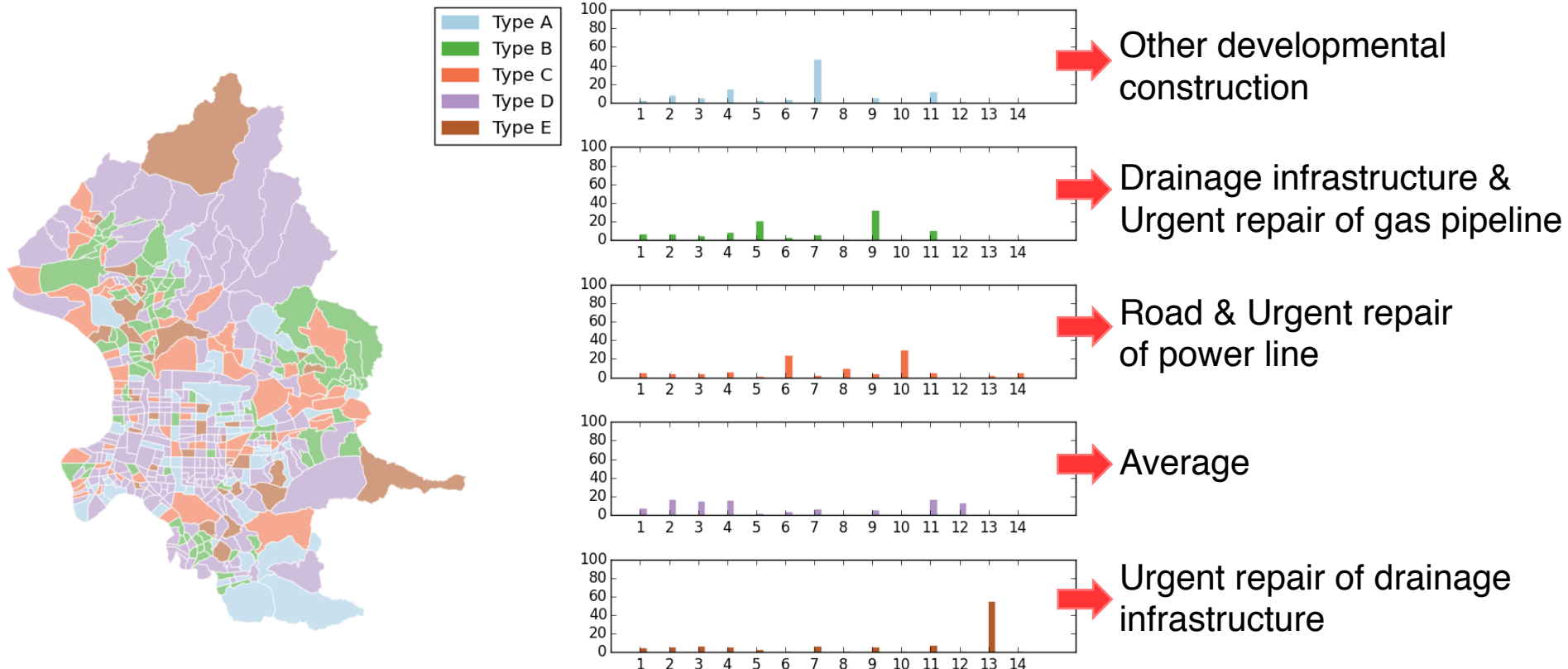
2. Re-labeling the “construction types” N = 19865

Type	Major object of construction	# records
1	Gas pipeline	1111 (5.59%)
2	Powerline	3388 (17.06%)
3	Water pipeline	3660 (18.42%)
4	Telecom cable	4138 (20.83%)
5	Drainage infrastructure	738 (3.72%)
6	Road	773 (3.89%)
7	Other developmental construction	792 (3.99%)
8	Non-governmental construction	18 (0.09%)
9	Urgent repair of gas pipeline	936 (4.71%)
10	Urgent repair of powerline	65 (0.33%)
11	Urgent repair of water pipeline	3894 (19.60%)
12	Urgent repair of telecom cable	210 (1.06%)
13	Urgent repair of drainage infrastructure	93 (0.47%)
14	Urgent repair of other objects	4 (0.02%)

Clustering-based Region Exploration

- Method: unsupervised K-means clustering algorithm

14 labels K-means: 5 clusters



Regional Predictive Analysis

Step 1. Autoregressive distributed-lag model

$$Y(t) \sim \exp(\theta(t))$$

$$\theta(t) = \alpha + \varphi_1 \cdot \theta(t-1) + \varphi_2 \cdot \theta(t-2) + \varphi_3 \cdot \theta(t-3) + \hat{\beta} \cdot \hat{x}$$

Step 2. Regional features

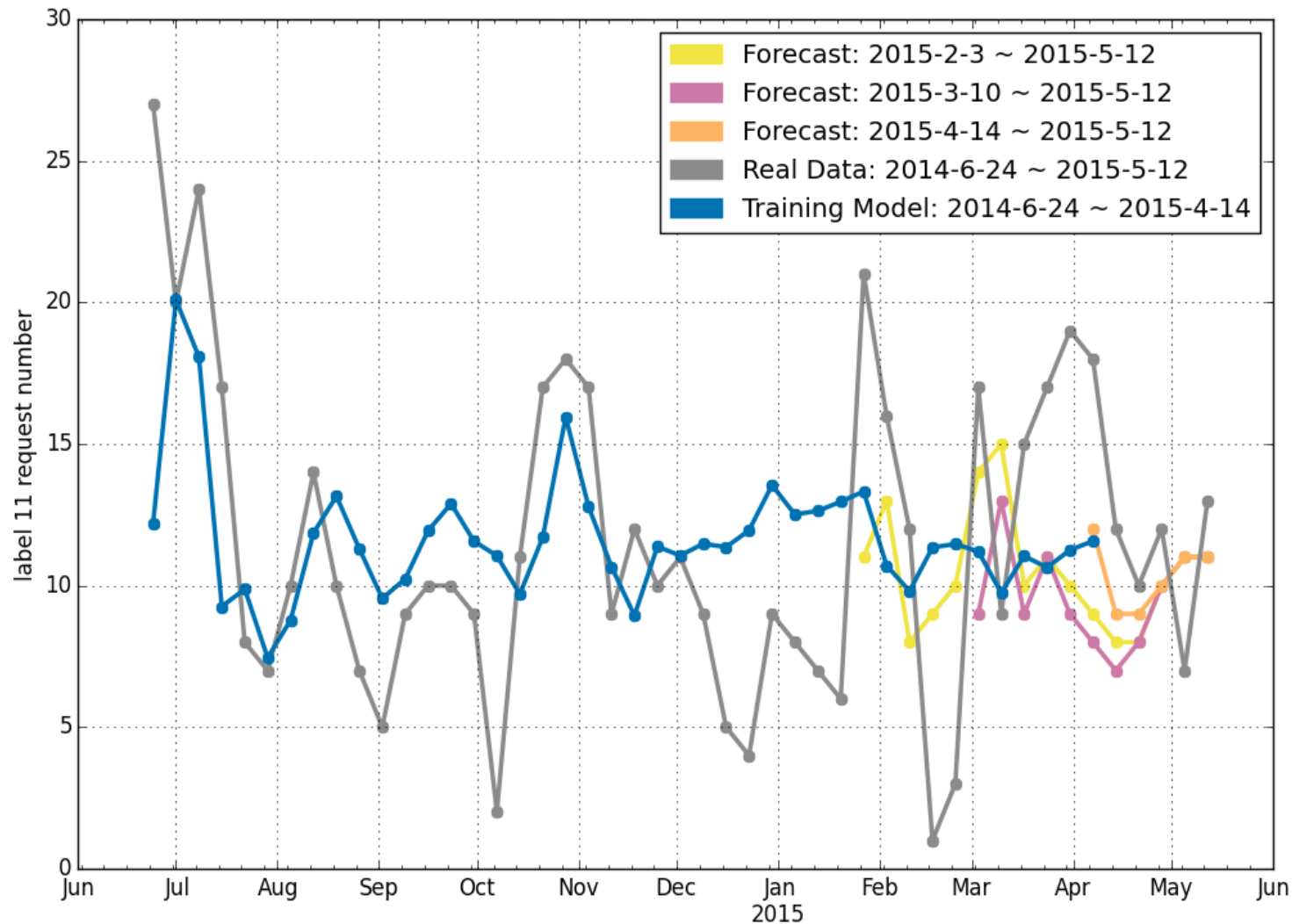
$$\hat{x} = \begin{bmatrix} \text{number of bus stations} \\ \text{number of MRT stations} \\ \text{area of historic waterflooding regions} \\ \text{district population} \end{bmatrix}$$

Table 2: The parameter values learned in our model.

Parameter	Coefficient	p-value	Description
α	0.0543	0.000	Intercept or global mean
φ_1	0.0264	0.000	Number reports previous week
φ_2	-0.0025	0.743	Number reports two weeks before
φ_3	-0.0259	0.001	Number reports three weeks before
$\hat{\beta}_1$	0.1043	0.000	Standardized number of bus stations
$\hat{\beta}_2$	0.2185	0.000	Standardized number of Metro stations
$\hat{\beta}_3$	0.2610	0.000	Standardized number of population
$\hat{\beta}_4$	0.0472	0.000	Standardized area of water-drone regions

Regional Predictive Analysis (cont.)

Step 3. Poisson regression and the predicted number time series



Regional Predictive Analysis (cont.)

Step 4. Model validation: rolling forecasting origin

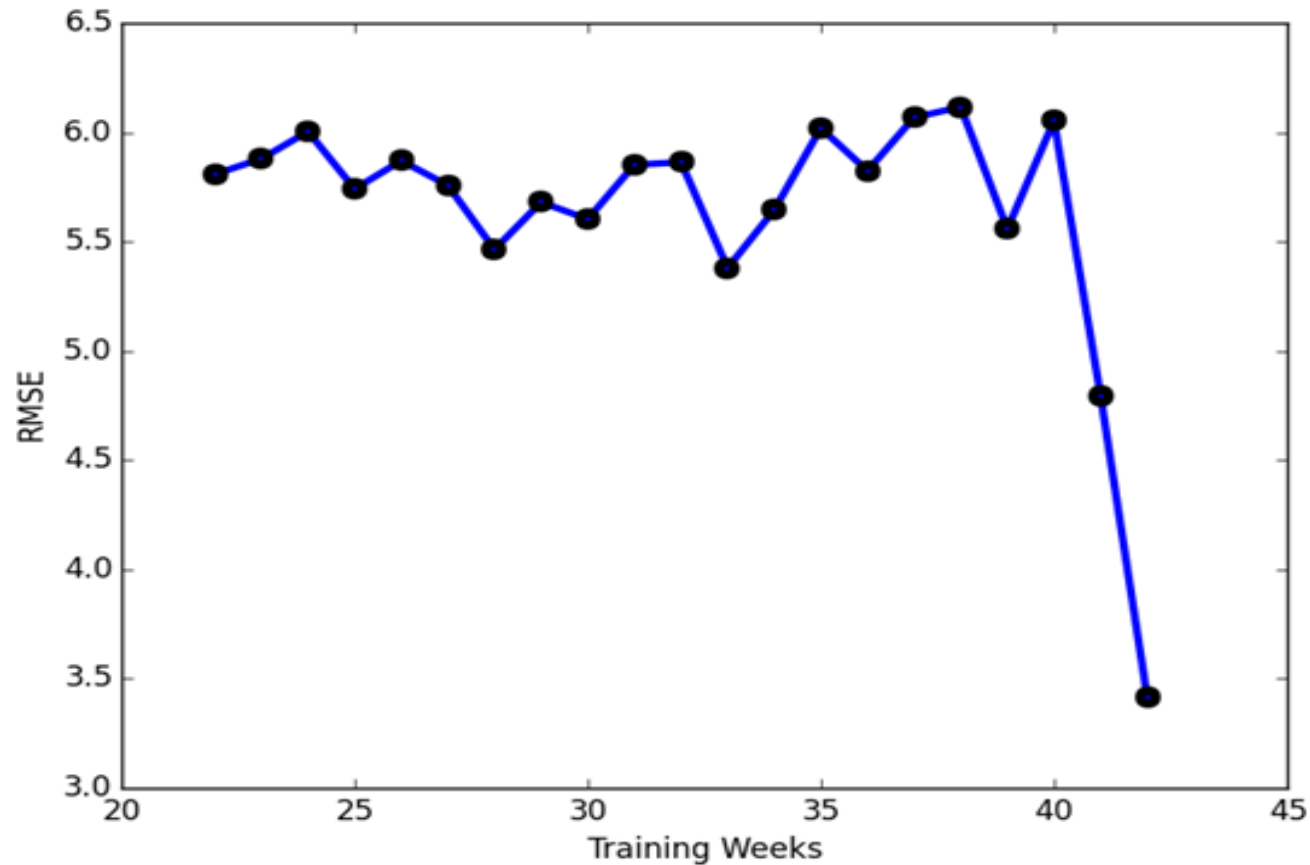


Figure 5: The RMSE values of the forecasted under our model.

System Demonstration

- Website: <https://junipertcy.info/urbcomp/index.html>

Summary

We proposed a **framework** to present, analyse, and predict urban construction data, featuring:

- (i) interactive visualisation
- (ii) clustering of similar request patterns
- (iii) regional request number prediction

Revealing the hidden information

→ **initiate** new research topics

Generalization

other data with **similar schema**

Prospects

- [*Academic-side*] Accuracy
 - Better algorithms
 - Validation of the result
- [*System-side*] Generalization
 - Including not only the construction data
- [*Practical-side*] Application
 - Bridging with our government
 - Building the system backend

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